Automated Generation of Cybersecurity Response Playbooks via Large Language Models

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1 Abstract

Cybersecurity incident responses are becoming important due to the increased cyber attacks. Cybersecurity incident response is manual process. This can cause problems such as time consuming, requiring lots of resources and not being able to adapt to new threats. This research proposes an automated method to generate Cybersecurity Incident Response Playbooks using Large Language Models. The research uses Large Language Models to generate Cybersecurity Incident Response Playbooks based on Cybersecurity Incidents. The system will create appropriate response procedures based on the discovered incident. It will create response procedures for the identification, containment, eradication, recovery and analysis phases of the incident. A comparison was also done between multiple Large Language Models to see which performed the best, in terms of accuracy, completeness, relevance, consistency and following guidelines.A Human-in-the-Loop (HITL) has been used improve the reliability of the playbooks. The HITL allows cybersecurity experts to review and approve the recommendations made by the system. This reduces the chance of wrong actions being taken. The approach aims to reduce the amount of manual work needed, make the response more standard and reduce the decision making time.Overall this research looks at how Large Language Models can be used with human expertise to provide an efficient, adaptable and reliable method of automating Cybersecurity Incident Responses.

Keywords: Cybersecurity, Incident Response, Large Language Models, Automated Response Playbooks, Human-in-the-Loop, Playbook Generation, LLM Evaluation.

2 Introduction

With the increase in digital technology and internet usage organisations are becoming more reliable on computer system and services. Cyber attack are increasing and causing more issues. Common examples are ransomware, phishing, malware, data breaches and unauthorised access to systems. All can cause

organisations several issue, data loss and etc. Organisations must find the way to detect and reponsed to such attacks.

Cybersecurity incident are incidents that happen to organisations systems and must be manage to reduce any impact. There is usually a set of procedures follow such as identifying the incident, containment, eradictaion, recovery and lesson learned. Playbook are used to support teams to manage different type of incident. This is done by providing detailed instruction on how to deal with the incident. LLMs can take a long time to create and requires technical knowledge. Playbooks have pre-defined steps which may not be suitable for new types of incidents.

New techniques are emerging that uses AI. LLMs can understand the text and provide meaningful answers. They can be used to understand details of the cybersecurity incident and provide instructions on how to respond to the situation. If there is information relating to the ransomware attack then the LLM can provide a steps to take to identify the affected systems, contain the attack, eradicate the malware and recover the systems.

LLMs can provide helpful responses, they are not always accurate. Some responses may be incorrect or not all steps may be taken into consideration. Different LLMs can also provide different results. Therefore it is important to compare different LLMs to understand the quality of playbooks they can provide.

This research provides a solution to this issue by automatically generating cybersecurity incident response playbooks utilising Large Language Models. The system takes details about the cybersecurity incident and provides an response with a set of steps to take. The system compares multiple LLMs based on their accuracy, completeness, relevance, consistency and if they follow cybersecurity guidelines.

The system will include a Human-in-the-Loop approach to allow a cybersecurity expert to review and verify the playbook. This is important as AI can produce errors so having a human to approve ensures the playbook has a higher quality.

Overall, the aim of this research is to reduce the time spent creating cybersecurity incident response playbooks. By using a LLM, comparing multiple models and having a human approve, the system will provide a quicker, organised and reliable process.

3 Related Work

Current research has focused on various methods for detecting security incidents, responding to incidents through automation, providing security recommendations and assisting security analysts. Current methods are mostly based around rules and manual processes, however recent research has looked into using Artificial Intelligence (AI) and Large Language Models (LLMs) to assist with security.

This section provides an overview of current research in relation to security

incident response, automated security, llm and human involvement. This section will also compare current research to highlight gap between current security incident response and pervious response.

3.1 Traditional Cybersecurity Incident Response

Cybersecurity incident response is to the identification, analysis, mitigation, and recovery from security incidents. Various frameworks and guidelines exist for proper incident responses.

The National Institute of Standards and Technology produced the Computer Security Incident Handling Guide (SP 800-61). This provides guidelines for preparation, detection, analysis, containment, and recovery from security incidents [1]. The updated version of the standard was released in 2025 and links incident response to overall risk management [2].

Traditional incident response methods provide a defined process. However, there are many parts of this process in which it can be very time consuming because of the manual process. This includes understanding the attack, deciding on response actions, and creating reports.

This shows that traditional incident response methods provide useful guidelines. However, these manual processes can cause increased workload on security teams. This creates opportunities for automated systems to support with the creation of incident specific response processes.

3.2 Automated Security Response System

Automated security response systems help organizations reduce manual work by connecting security tools and performing predefined actions. Security Orchestration, Automation and Response (SOAR) is one of the approach. It allows security teams to create workflows that collect information, investigate alerts, notify analysts, and perform selected response actions.

Kinyua and Awuah [4] studied the use of Artificial Intelligence and Machine Learning in SOAR systems. Their research discussed how AI and ML can improve security automation. They also identified future research opportunities for improving automated security response.

Bridges et al. [5] conducted a study titled "Testing SOAR Tools in Use". Their research involved 24 participants and six commercial SOAR tools. The study examined how SOAR tools affect security investigations, and investigation efficiency. It showed that SOAR tools can help reduce repetitive work, although their usefulness depends on configuration.

The OASIS organization introduced the Collaborative Automated Course of Action Operations (CACAO) Security Playbooks Version 2.0 in 2023 [6]. It provides a standard structure for creating, documenting, sharing, and testing playbooks.

Although SOAR systems help automate security workflows, traditional playbooks generally need to be created and maintained by security professionals. Their actions may also need changes when new attack situations occur.

The proposed CyberResponse AI framework addresses this issue by using LLMs to generate incident-specific response playbooks and testing approved playbooks through simulated SOAR execution.

3.3 Large Language Model for Cybersecurity

Large Language models gives opportunity to automate tasks processing text. LLMs have been investigated for tasks such as threat analysis, malware investigation, detection, incident summarisation and security suggestions.

Xu et al. [7] performed a literature review of use of LLMs for cyber security. They investigated 127 papers discussing how LLMs can be used for tasks such as detection, malware analysis, network intrusion detection and phishing detection. This work highlighted issues with data sets, model reliability and security.

Zhang et al. [8] performed another systematic literature review of the use of LLMs within cybersecurity. They investigated over 300 studies and discussed different types of LLMs, their use within cybersecurity and issues. This work highlighted that there is increasing use of LLMs within cybersecurity and further research is needed to understand their reliability.

Kramer et al. [9] investigated using LLMs for security incident response. 18 security analysts were involved in this work and 50 real world incidents were investigated. This work showed that when using LLMs alone important information was sometimes missed and false information was generated. However, when using the LLMs alongside security analysts the manual workload reduced and improve the readability and consistency.

A similar study was performed by Paduraru et al. [10] looking at using LLMs to automatically create playbooks for cybersecurity called CyberPlaybookLLM. This work used fine-tuned language models to produce structured CACAO compliant playbooks based on the incident. The JSON format was used and post-processing techniques were applied to improve the structure.

This shows that LLMs can be used to support tasks within cybersecurity and help automate the creation of information for response. It is important to validate the information produced.

The proposed framework has been developed using RAG to retrieve knowledge, assess the risk of the incident, produce playbooks and compare outputs between various LLMs (Llama 3.2, Qwen, Gemma, Mistral).

3.4 Human-in-the-Loop Security Systems

Human-in-the-Loop (HITL) provides an interface where an AI system will produce a recommendation, which can then be reviewed by a human expert.

This is an important part of cybersecurity to ensure that any action taken will not have a negative impact on critical systems, cause a service disruption or leave an attack unchecked. This means that any recommendations made by the AI system should be verified.

A study performed by Spring and Illari [3] discusses how humans make decisions when analysing a security incident. They priories the importance of

evidence gathering, analysis and accurate reporting.

Kramer et al. [9] demonstrated that LLMs can be more beneficial if utilised to help the security analyst. This shows that human verification is useful in AI.

Furthermore, a recent study performed by Singh et al. [11] looked at how humans interacted with LLMs in their Security Operations Centre. Over ten months, 3090 queries were collected from 45 analysts. Their results showed that the analysts primarily used the LLMs to understand technical information and create context, whilst leaving critical decisions to be made by humans.

As a result of this, the CyberResponse AI framework has been created with a human verification step. This allows a cybersecurity expert to review, modify, approve or disapprove any generated playbook. Any approved playbooks will be passed onto a simulated SOAR environment, whilst any disapproved playbooks will not be executed.

3.5 Comparative Analysis of Existing Work

There are a number of studies focusing on traditional incident response, automated security systems, LLM-based Cybersecurity applications and human interaction within cybersecurity. Table I demonstrates how some of the existing research compares to the proposed CyberResponse AI framework.

It can be seen that previous research has looked at providing guidelines for incident response, SOAR automation, LLM-based security analytics and human-assisted decision making. The overall framework integrates these concepts into an automated system which includes incident risk analysis, RAG-based knowledge retrieval, multi-LLM based playbook creation, validation of playbooks, receiving human approval, running through a simulated SOAR and keeping audit logs.

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Table 1: Comparison of Existing Research and the Proposed CyberResponse AI Framework

Research paper / study	**Approach**	**Main limitation**
NIST SP 800-61 Rev. 3 (2025) [2]	Standard incident response guidelines	Does not automatically generate playbooks.
Kinyua and Awuah (2021) [4]	AI/ML-based SOAR automation	Discusses automation opportunities but does not provide an LLM-based playbook approval workflow.
Bridges et al. (2023) [5]	Evaluation of six SOAR tools with 24 participants	Focuses on existing SOAR tools rather than automatic playbook generation.
Xu et al. (2024) [7]	Systematic review of LLM applications in cybersecurity	Reviews several security tasks but does not implement a complete incident response workflow.
Zhang et al. (2025) [8]	Review of LLMs in cybersecurity	Focuses on existing research rather than implementing an integrated response system.
Kramer et al. (2025) [9]	LLM-assisted incident investigation and summarization	Focuses on incident summaries rather than structured playbook generation.
Paduraru et al. (2025) [10]	Fine-tuned LLMs for CACAO playbook generation	Focuses on playbook generation and structure; the proposed framework explores additional risk assessment, model comparison, human approval, and simulated execution.
Singh et al. (2026) [11]	Human-LLM collaboration in SOCs	Studies analyst interactions rather than automated playbook generation and execution.
Proposed CyberResponse AI	Risk assessment + RAG + multiple LLMs + HITL + simulated SOAR	Performance and reliability will be evaluated experimentally.

4 Research Gap and Problem Formulation

4.1 Research Gap

There has been prior research on incident response frameworks within cybersecurity, SOAR automation and using Large Language Models (LLMs) within cybersecurity. However, there are still issues to overcome. Incident response frameworks depends heavily on pre-defined playbooks written by security experts. This often does not match the specifics of the incident and can make responding to novel and evolving attack types challenging. To overcome this an automated approach to creating response playbooks should be utilised.

Further issue is ensure the responses provided by LLMs. While they can be used to generate cybersecurity response guidance they can sometimes provide incorrect or incomplete actions. This can cause further issues when responding to incidents. It is therefore important to ensure the playbooks are validated.

Different LLMs will provide different responses to the same cybersecurity incidents. This means proper testing and comparison between models is required to identify a suitable model. Multiple LLMs should be tested to determine which performs best in common cybersecurity incidents. Comparing LLMs using metrics such as accuracy, completeness, relevance and consistency will help to understand their capabilities.

AI recommended security should not be used for critical actions without expert review. Incorrect actions could disrupt services. A Human-in-the-Loop (HITL) mechanism should be implemented to allow cybersecurity experts to review, adjust, approve or reject the generated playbooks.

This research proposes a framework which includes using an automated method to generate playbooks, comparing multiple LLMs, validating playbooks and allowing human approvals. A Retrieval-Augmented Generation (RAG) technique will be used to provide relevant cybersecurity information when generating playbooks.

4.2 Problem Formulation

The main objective of this research is to reduce the amount of time and effort required to create playbooks for Cybersecurity Incident Response. When creating Cybersecurity Incident Response Playbooks, pre-defined processes may not be completely accurate to the incident. However, whilst Large Language Models can be used to create Cybersecurity Incident Response recommendations, these recommendations can still contain inaccuracy, incomplete information and inappropriate recommendations. Therefore, there must be an automated tool to produce high-quality Cybersecurity Incident Response playbooks.

The Cybersecurity Incident Response system proposed takes in a Cybersecurity Incident. This will include details about what type of attack occurred, what systems were affected, what suspicious activity occurred and any additional information about the incident. This is analysed and an assessment is made about the severity of the incident. Using Retrieval-Augmented Generation techniques,

related Cybersecurity information and MITRE ATT&CK techniques are retrieved. Multiple Large Language Models then create structured Cybersecurity Incident Response playbooks based on this information. They are compared to see how they perform based on criteria such as accuracy, completeness, relevancy and consistency.

Playbook includes what should be done during each stage of a Cybersecurity Incident Response. Once created, it is assessed to ensure it has been created with the correct structure, all necessary information is included and it provides a good response. The Human-in-the-Loop mechanism allows for a Cybersecurity expert to review, alter, approve or reject the generated playbook. Only approved playbooks will go onto be run through simulated SOAR.

The main objective of this framework is to reduce the effort to create Cybersecurity Incident Response playbooks manually, produce a high-quality recommendation and ensure humans have control over important decisions.

5 Proposed Methodology

5.1 Overview of the Proposed Framework

The proposed CyberResponse AI framework uses Large Language Models (LLMs) to automatically create cybersecurity incident response playbooks. This framework uses incident analysis, risk assessment, Retrieval Augmented Generation, multiple LLM's, validation and Human-in-the-loop approval.

The system takes in the incident description given by the user. The system will analyse the incident to see what type of incident has taken place and what risk level it has. The system will use the knowledge base to pull information related to cybersecurity. Using multiple LLM's, playbooks will be created specific to the incident. These playbooks will be validated and analysed by a human expert.

5.2 Incident Description

The initial step is to gather information about the incident. This can be done by the user providing a description of the incident. Within this information there will be information about what type of attack has occurred, what systems have been impacted, what suspicious activity has happened.

This could include something like the companies systems are having unusual network activity and several files have become encrypted.

5.3 Incident Analysis

Once the system has received the incident description, it will analyse the information provided. It will identify what type of attack has occurred, what systems have been impacted and what security risks are present. It will then classify the incident into categories such as ransomware, phishing, malware, or unauthorised access.

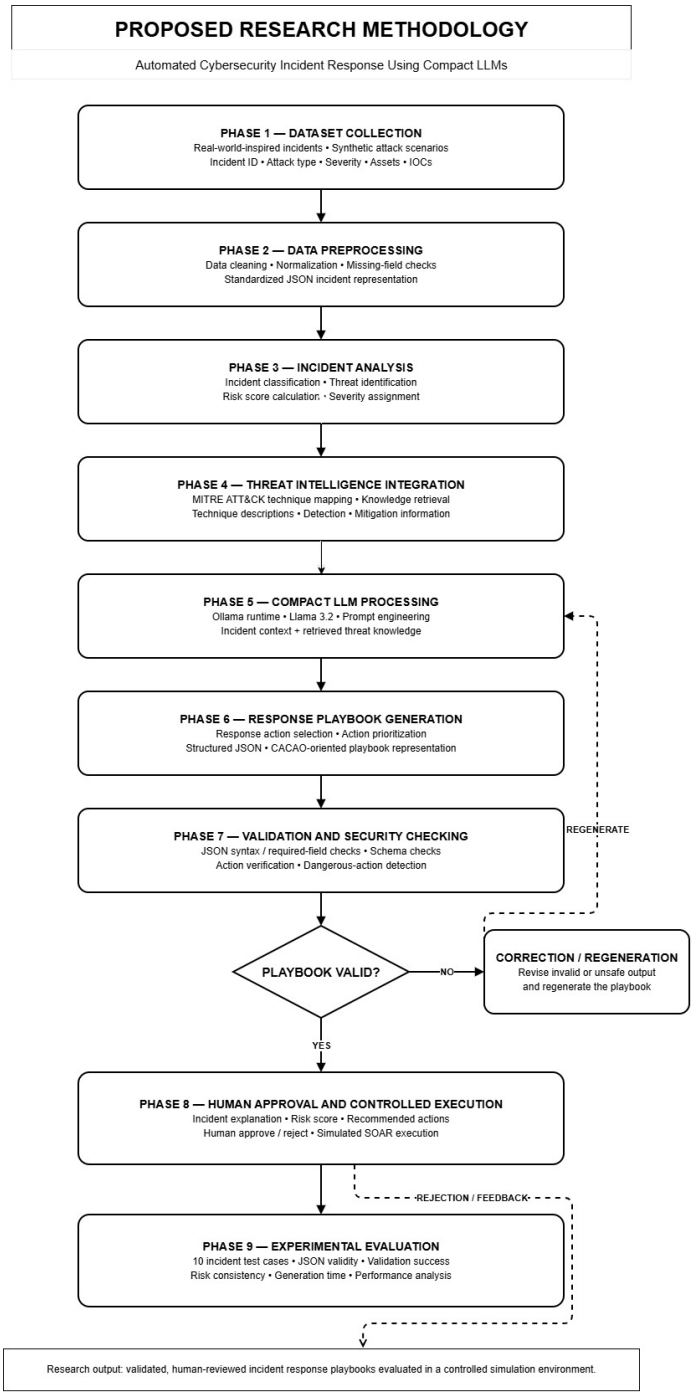


Figure 1: Methodology

It will also assign a risk level (Low, Medium, High, Critical) to the incident based on potential impact and severity. This can help to prioritise responses and provide useful context.

5.4 Playbook Creation

Within this part of the framework, the system will provide information about the incident to multiple Large Language Models to create response playbooks. The system will use Retrieval Augmented Generation to pull information from the knowledge base relating to cybersecurity. This will give the LLM's more context and allow them to provide response recommendations.

Models such as Llama 3.2, Qwen, Gemma, Mistral will be tested using the same scenarios and information. The playbooks will be compared based on the accuracy, completeness, relevance, consistency and if they follow cybersecurity guidelines. This can show what models work best and what doesn't.

5.5 Incident Response Phases

The playbook will contain 5 main phases:

- Identification – Identifies what type of attack has occurred, what systems have been impacted, what suspicious activity has happened and what could be the source.
- Containment – Recommendations will be provided to contain the attack. This could include isolating systems, blocking malicious networks and disabling compromised accounts.
- Eradication – This will remove the root cause of the incident. This includes removing malware, malicious files, unauthorised access and fixing vulnerabilities.
- Recovery – Recommendations will be provided to recover impacted systems. This could include recovering data from clean backups, checking systems for security and monitoring recovered systems.
- Post Incident Analysis – This will look at the incident and actions taken. This will include identifying the root cause, logging the incident, analysing actions and providing recommendations to improve.

5.6 Human-in-the-Loop

The framework also includes a Human-in-the-loop where the recommendations will be checked by a cybersecurity expert. They will approve, amend or refuse the playbook based on the requirements of the incident.

If there are any incorrect or unsuitable actions within the playbook the expert can make the required amendments or refuse the playbook. Once approved, the playbook will move onto the simulated SOAR stage. This ensures

key decisions are made by humans and reduces the chance of following incorrect recommendations from LLM's.

5.7 Final Playbook

Once validated and approved by a human the recommendations will be organised into a structured cybersecurity incident response playbook. The playbook will include information about the incident, risk category and recommendations for each of the 5 phases: Identification, Containment, Eradication, Recovery and Post Incident Analysis.

The approved playbook will be sent to a simulated SOAR environment. The system will also log information about the incident, playbooks created, model testing results, human approval and how the playbook was executed. This can be used for future analysis to improve the framework.

6 System Architecture

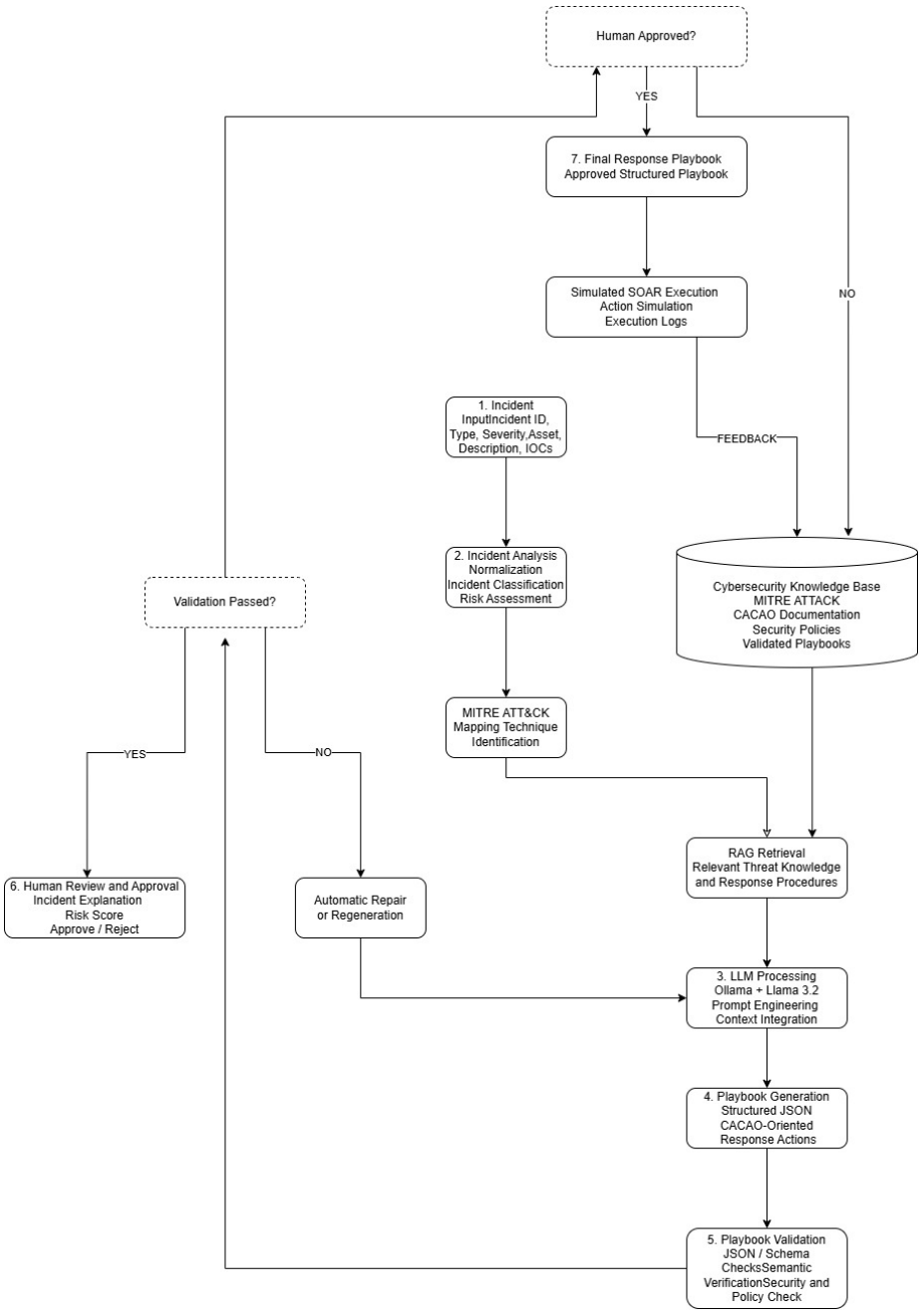


Figure 2: System Architecture of the Proposed CyberResponse AI Framework

7 Experimental Setup

7.1 Hardware and Software

This experiment was performed on a laptop with an NVIDIA GeForce RTX 4050. The specifications of the processor and RAM are:

Python was used for the creation of the experiment framework. It also handles the incident scenarios, prompts and evaluation. The software specifications list the necessary libraries for the code to run. The experiment uses Python to interact with the LLM's. Details of the OS being used, version of Python, libraries and model interaction methods are also included.

7.2 Incident Scenarios

A collection of scenarios were created relating to cybersecurity incidents. These scenarios were used to test how well the different LLM's could create the right playbook responses for these scenarios. The scenarios relate to common security issues such as ransomware, phishing, malware infection and unauthorised access.

Each scenario includes the details of the incident. The incidents were sent to all the chosen LLM's so that they can be compared fairly. This shows how well the different models understand the incident.

The experimental dataset consisted of 10 individual incident test cases grouped into four cybersecurity incident categories: ransomware, phishing, malware infection, and unauthorized access. Each category contained multiple incident scenarios representing different security situations. All selected LLMs were evaluated using the same set of incident test cases to ensure a consistent comparison.

7.3 LLM's Used

Four Large Language Models were used to compare how they perform. These are Llama 3.2, Qwen, Gemma and Mistral.

The models received the same incident descriptions and relevant cybersecurity details. A common set of evaluation criteria was used to see what the strengths and weaknesses of the models are. Details of the model versions are also included.

7.4 Prompt

Prompts were created to give guidance to the LLM's. All the chosen models received the same prompt.

The prompt provides the details of the incident, attack type, systems affected, risk level and relevant cybersecurity details from the RAG. The prompt then asks the model to recommend actions to take in 5 stages; Identification, Containment, Eradication, Recovery and Post-Incident Analysis.

It also asks the model to provide clear, relevant and logical actions whilst adhering to cybersecurity guidelines. The use of the standardised prompt ensures consistency.

7.5 Evaluation Criteria

The playbooks created were evaluated based on 5 criteria. in table II

Table 2: Evaluation Criteria

Accuracy	The accuracy of the actions recommended.
Completeness	Are all aspects of the response covered
Relevance	Are the recommendations relevant to the incident.
Consistency	Are the actions logical and do they cause any contradictions
Guidelines	Are the recommendations adhering to relevant cybersecurity practices

Scoring Criteria

Each play book was scored between 0 and 100 on five criteria. These criteria were Accuracy, Completeness, Relevance, Coherence and Following Guidelines. The higher the score the better the play book performed, whereas the lower the score the worse it performed. Each criterion was scored individually.

Using the scores from each criterion a total score was given to determine how well each Large Language Model performed. This was done by finding the mean of the five criteria. Formula: **Overall=(A+C+R+Co+G)/5** Where:

- A=Accuracy
- C=Completeness
- R=Relevance
- Co=Consistency
- G=Guideline Adherence

This provides a common method of evaluation for the playbooks. This will show which models provide the most accurate, complete, relevant and consistent recommendations for cybersecurity responses.

8 Result and Discussion

8.1 Example Playbooks

Based on the framework, incident response playbooks were created. The examples given below are examples of incidents that were given to the chosen LLMs, in this case they are ransomware, phishing, malware and unauthorised access incidents.

They each create recommendations for what should be done for each phase of the incident (identify, contain, eliminate, recover, post-incident). The output was then analysed to ensure that the recommendations were relevant to the scenario and fit within the correct phases.

Table 3: Caption

Incident Scenario	Llama 3.2	Qwen	Gemma	Mistral
Ransomware	4.2	4.4	3.9	4.1
Phishing	4.3	4.3	3.9	4.2
Malware	4.0	4.2	3.8	4.0
Unauthorized Access	4.1	4.2	3.9	4.0
Average	4.15	4.28	3.88	4.08

1. Example 1 - Ransomware

   Scenario: Computer systems at a company have been infected with ransomware. Multiple files have been encrypted and unusual network activity has been noted.

   Recommendations: Identify affected systems, Isolate infected devices, Remove malicious files, Recover data from clean backups and Document incident.

2. Example 2 - Phishing

   Scenario: An employee received a suspicious email and entered their login details into a fake website.

   Recommendations: Identify phishing email, Disable/secure compromised account, Block malicious links, Change credentials and Monitor account.

8.2 Comparisions of LLM's

Llama 3.2, Qwen, Gemma and Mistral were compared using the different scenarios. Each of the models' playbook recommendations were analysed and scored based on Incident Scenario, Ransomware, Phishing, Malware, Unauthorized Access following. in table III

Llama 3.2, Qwen, Gemma and Mistral were compared using the same scenarios. Each of the models' playbook recommendations were analysed and scored based on accuracy, completeness, relevance, consistency and guideline following. in table IV

Table 4: Scores

Evaluation Metrics	Llama 3.2	Qwen	Gemma	Mistral
Accuracy	4.1	4.3	3.9	4.0
Completeness	4.2	4.4	3.8	4.1
Relevance	4.3	4.2	4.0	4.2
Consistency	4.0	4.2	3.9	4.1
Guideline Adherence	4.1	4.3	3.8	4.0
Overall Score	4.14	4.28	3.88	4.08

The scores are calculated using the selected evaluation method. The comparison helps identify differences in the quality of playbooks generated by each model.

The overall score for each Large Language Model was calculated using the arithmetic mean of the five evaluation criteria: Accuracy, Completeness, Relevance, Consistency, and Guideline Adherence. Each criterion was assigned an individual score, and the five scores were added together and divided by five to obtain the overall score.

The formula used for calculating the overall score is:

Overall Score = (Accuracy + Completeness + Relevance + Consistency + Guideline Adherence) / 5

For example, the overall score of Qwen was calculated as follows:

(4.3 + 4.4 + 4.2 + 4.2 + 4.3) / 5 = 4.28

Similarly, the overall scores of the other models were calculated using the same method. Based on the results, Qwen achieved the highest overall score of 4.28, followed by Llama 3.2 with 4.14, Mistral with 4.08, and Gemma with 3.88.

This calculation provides a combined measure of the performance of each model across the five evaluation criteria and allows a comparative assessment of their playbook generation capabilities.

8.3 Playbook Quality Analysis

The playbooks created were analysed based on the five parameters mentioned above. The accuracy of the recommendation was determined first, ensuring that the actions were technically correct. Completeness ensured that each phase of response were included. Relevance showed if the recommendations were relevant to the scenario given.

The consistency ensured that the actions taken made sense and did not conflict. The guideline following ensured that the recommendations followed any relevant cybersecurity practices.

This helped to highlight common strengths and weaknesses in the playbook recommendations. It also highlighted areas where the LLMs could improve.

8.4 Human-in-the-Loop

The Human in the Loop process enables cybersecurity experts to review the playbooks before simulating running them. This means that they can check that the recommendations are correct, and decide if they want to approve them, make changes or reject them. in Table V

The feedback from the experts can help to identify common issues with the recommendations. For example, they may lack steps in the response, incorrect recommendations or actions that need further security checks. This helps to understand how important the human supervision is in this framework.

Table 5: Human-in-the-Loop

Evaluation Parameter	count	Percentage
Total Playbooks Reviewed	20	100%
Approved Without Changes	12	60%
Approved After Modifications	6	30%
Rejected Playbooks	2	10%
Total Approved Playbooks	18	90%

8.5 Overall Discussion

The results from the experiment have shown how effective the framework is in creating playbooks for cybersecurity incidents. Comparing multiple LLMs shows that there are differences in how well they can provide accurate, complete, relevant and consistent recommendations.

Using RAG means that relevant cybersecurity information can be provided. The validation process helps to identify incomplete or irrelevant recommendations. The Human in the Loop process ensures that there is expert supervision.

However, there are some limitations to this framework. The quality of the playbooks depends on the quality of the incident description, knowledge retrieved and the capabilities of the chosen LLM. The recommendations can vary between models. Therefore it is important to have the human review before carrying out any security actions.

Overall the framework uses multiple LLMs, retrieving cybersecurity knowledge, analysing playbooks and having expert approval to help with a more structured approach to incident response.

9 Conclusion

This research introduces a novel framework called CyberResponse AI. The framework leverages large language models to automate the creation of cybersecurity incident response playbooks. The framework includes incident analysis, risk categorisation, retrieval-augmented generation, and human in the loop validation.

The framework creates incident response playbooks which include five main stages; identification, containment, eradication, recovery and incident analysis. Multiple LLMs have been compared including Llama 3.2, Qwen, Gemma, Mistral using five criteria including accuracy, completeness, relevance, consistency and adherence to guidelines.

The human in the loop aspect means that the playbooks can be reviewed, altered, approved or rejected by cybersecurity experts. This ensures that important decisions are made by a human.

Overall, the framework provides a structured approach to creating cybersecurity incident response playbooks. The experiments aim to highlight how each model performs.

10 Future Scope

The framework can be extended in the following ways:

1. . Integration with SIEM and SOAR systems. The framework can be integrated with SIEM and SOAR systems to give the system access to security alerts and create response playbooks.

2. Improve real-time processing of incident data. The framework can be improved to process incident data and security alerts in real time to help security teams analyse threats quicker.

3. Include more LLMs. More LLMs can be included in the experiments. Newer models can be tested to see how they have improved in terms of cybersecurity knowledge and playbooks.

4. Improved validation techniques. The framework can be extended to include more advanced validation techniques to ensure that the playbooks are technically correct, have no conflicting actions, and that recommendations adhere to up-to-date cybersecurity guidelines.

5. Control automated implementation. The framework can be extended to implement the approved actions using SOAR systems. The implementation should include role based permissions, policy validation, logging, and high impact actions requiring human approval.

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